

Kenya AI Challenge - AgriFin Track Submission

1. Team Name and Members

Team Name: AgriLegends

Team Members: Karen Miano & Allan Kariuki

2. Problem Route

Official Kenya AI Challenge AgriFin brief.

3. Selected Brief

Farm Mall Ventures Ltd - Satellite-Based AI Monitoring for Potato Productivity and Market Confidence.

4. Problem Understanding

Smallholder potato farmers across Kenya lack continuous, objective field visibility after planting, creating both an agronomic and a commercial blind spot. Agronomically, potatoes are highly vulnerable to fast-onset stress such as pests, disease, moisture and nutrient deficiency, but occasional visual checks mean damage is usually caught only after it's irreversible beneath the canopy, costing yield and tuber quality. Commercially, that same blindness breaks trust: lenders, off-takers, and aggregators have no objective way to verify crop health, forecast output or confirm production history, so they price smallholders as high-risk and exclude them from premium contracts, pre-harvest commitments and financing. The core problem is a lack of localized, actionable intelligence, which strips farmers of commercial leverage.

5. Target User and Pain Point

- Primary User: Smallholder potato farmers in Kenya who lack timely tools for proactive management and are therefore locked out of potential value chains due to unverified production.
- Secondary User: off-takers, potato processors, aggregators, buyers and financial partners.
- Exact Pain Point: Farmers currently rely on infrequent, manual physical inspections to monitor crop health. Because they lack proactive monitoring tools, critical crop stressors are only identified after the damage is done, this latency leads to reduced yields and the agronomic data deficit causes commercial exclusion because farmers cannot generate empirical verifiable records of their crop's health or project an accurate harvest volume. Warning before a stress event becomes a yield loss only discovered at harvest.

6. Evidence or Research

- KEPHIS/NPCK: certified seed meets <5% of demand; agriculture is ~25% of GDP, ~65% of export earnings confirms scale and root cause.
- NPCK has publicly flagged the absence of standardized digital traceability as a barrier to premium buyers (e.g. processors with strict specs, and is only now piloting traceability; the gap is real and unaddressed.
- Apollo Agriculture (Kenya, since 2016) proves the model at scale: satellite + soil + behavioral data feeding an ML credit-risk engine has reached 350,000+ smallholders with yields 2-2.6x the national average strong evidence this is viable, not speculative.

- KAMIS (market prices) and KIAMIS (farmer registry) exist but expose no public API confirmed by direct inspection so the market-confidence and identity layers are a genuine integration gap.
- ODPC's 2024 Digital Credit Provider guidance and CBK's 2022 Digital Credit Providers Regulations confirm an active regulatory framework this solution must design against from day one.

7. Proposed Solution

A farmer registers a plot by confirming its boundary. From there, the solution runs as three connected layers:

- Ingestion layer (cumulative data pipeline): continuously pulls satellite vegetation indices (NDVI/EVI), soil baseline, and weather for that exact plot, no manual entry, and cross-checks this against farmer-submitted ground truth (planting dates, photos taken in intervals. Every pass adds to the plot's record; nothing is overwritten or reset each season.
- Processing layer: that cumulative stream is dimensionally and factually modeled in Neo4j, where vegetation-index computation, anomaly/stress detection, yield-forecast ML, and market-price normalization all run as queries over one connected graph instead of siloed tables. A detected stress signature (e.g., an unexpected NDVI dip vs. the plot's own history) triggers an alert in time to act, not after harvest, and the same graph produces a running harvest forecast (yield and timing) through the season.
- Reporting layer: farmers get real-time alerts and a mobile dashboard of crop, stress, weather, and harvest analytics; lenders and cooperatives get a compliance dashboard and credit-score risk view where every score traces back to the specific observations behind it, not a black box.

8. AI, Data, and Neo4j Role

- Stress detection: anomaly detection on NDVI/EVI/NDMI per plot vs. that plot's own historical baseline.
- Harvest forecasting: regression/gradient-boosted model combining vegetation-index trajectory, soil fertility class, rainfall, and planting date, recalculated on every satellite pass
- Neo4j is the system's core. Every farmer, plot, season, observation, soil sample, stress event, harvest, sale, and loan is a node, connected explicitly (Farmer-OWNS->Plot-HAS_SEASON->Season-YIELDED->Harvest-SOLD_VIA ->Transaction):
 1. GraphRAG advisory - natural-language queries translated to Cypher, so answers are grounded in recorded data, not general model knowledge.
 2. Explainable credit scoring - a lender traverses the graph to see exactly which observations produced a score, meeting the explainability bar ODPC's guidance effectively requires.
 3. Fraud/duplication detection - flags the same plot polygon or identity claimed across multiple farmer records, a classic graph use case relational schemas make far harder
 4. Portfolio/risk-network analysis - clusters farmers by agro-zone or shared exposure so a lender can see correlated risk instead of treating each loan as independent.

9. Build Plan

Build first: Neo4j AuraDB schema

(Farmer/Plot/Season/Observation/SoilSample/StressEvent/Harvest/Transaction) populated with

real data - Sentinel-2/OpenWeather Agro API NDVI-EVI series and live iSDAsoil soil properties for real coordinates in a Kenyan potato county farm.

Demonstrates: continuous ingestion into the graph - rule-based stress detection - dashboard showing an NDVI trend, an active alert, a simple harvest forecast, and an explainable mock credit score with its graph-traced evidence.

Working by deadline: real satellite, weather, crop, soil ingestion into Neo4j for several plots; a working stress-detection rule (rule-based, not full ML, given the timeline); a queryable explainable score; a basic web dashboard; a USSD sandbox flow (Africa's Talking) simulating a stress alert. Mocked/simulated: market-price and the credit score itself (rule-based, not trained on real repayment outcomes); AWS IoT sensors (if time allows)

10. Key Assumptions, Risks, and Responsible AI Use

- Resolution mismatch: Sentinel-2's 10m pixels may be coarse relative to sub-hectare smallholder plots; boundary accuracy and pixel-mixing are a real risk to stress-signal quality.
- Cloud cover: optical satellite passes can be blanked out for days during the long rains, risking false "no stress detected" readings; a production system would need Sentinel-1 radar as a cloud-penetrating fallback.
- Privacy/consent: plot location and transaction data are sensitive personal data under the Data Protection Act, 2019; explicit, specific consent and data-minimization are required before any data collection beyond the prototype.
- Fairness/bias: scoring inputs (soil fertility class, agro-zone) are geographic and largely outside a farmer's control; a model trained on these risks systematically penalises farmers in naturally lower-fertility zones regardless of farming skill. Mitigation: human loan-officer review on any adverse decision, never a fully automated rejection, in line with ODPC/CBK digital-lending guidance
- Trust: an opaque score will be rejected by farmers and loan officers alike — the explainability path through the graph is a trust requirement.

11. Support Needed

- Domain expert: an agronomist's input on potato-specific NDVI/NDMI stress thresholds by growth stage, since most public reference scripts are not potato-tuned.
- Data/partnership support: any existing contacts or pathways into KAMIS, KIAMIS, KMD, or KEPHIS data, since these are the largest unresolved dependencies for anything beyond a static-data demo.